

What Is a Subsystem?

THE BIG IDEA

Each subsystem owns exactly one mechanism and controls it completely.

One job, one file. If the scoop breaks, the drivetrain keeps driving.

KEY CONCEPTS

One job Separation

One mechanism.
One subsystem.
WPILib enforces it.

periodic() Sensors only

Runs every 20ms.
Reads sensors, logs.
No motor control.

Commands Motor control

Start, stop, interrupt.
Set a goal.
Scheduler controls it.

Goal pattern Coordination

Command sets goal.
periodic() reads it
and applies it.

Superstructure Preview

Coordinator above
multiple subsystems.
Built later.

BUILDING SCOOP (THE GOAL PATTERN)

- 1 Create the file with the WPILib tool (SubsystemBase) - never by hand
- 2 Add the hardware field (servo)
- 3 Add a Goal enum (FLAT, CARRY, DUMP)
- 4 Add @AutoLogOutput on the goal field
- 5 Add setGoal() and a setGoalCommand()
- 6 Wire periodic() to read the goal and apply it
- 7 Register in RobotContainer, bind a button, build + simulate + verify

AFTER THIS LESSON I CAN...

- ☐ Explain why periodic() cannot contain motor control
- ☐ Name the four parts of every subsystem
- ☐ Build a subsystem using the goal pattern
- ☐ Use the aksubsystem snippet to scaffold a new subsystem
- ☐ Spot all four Broken Robot Lab bug types in unfamiliar code
- ☐ Explain what a superstructure is and why it isn't built yet

QUESTIONS I STILL HAVE

Write your questions here. Bring them next session.

KEY VOCABULARY

SubsystemBase	WPILib base class - extending it registers with the CommandScheduler automatically
Goal	The target state a subsystem moves toward; set by a command, read by periodic()
Superstructure	A coordinator above multiple subsystems for multi-mechanism moves
@AutoLogOutput	AdvantageKit annotation that automatically logs a field to AdvantageScope
Requirement	The subsystem(s) a command claims; scheduler blocks conflicting claims

Write anything here — surprises, connections, things to look up later.